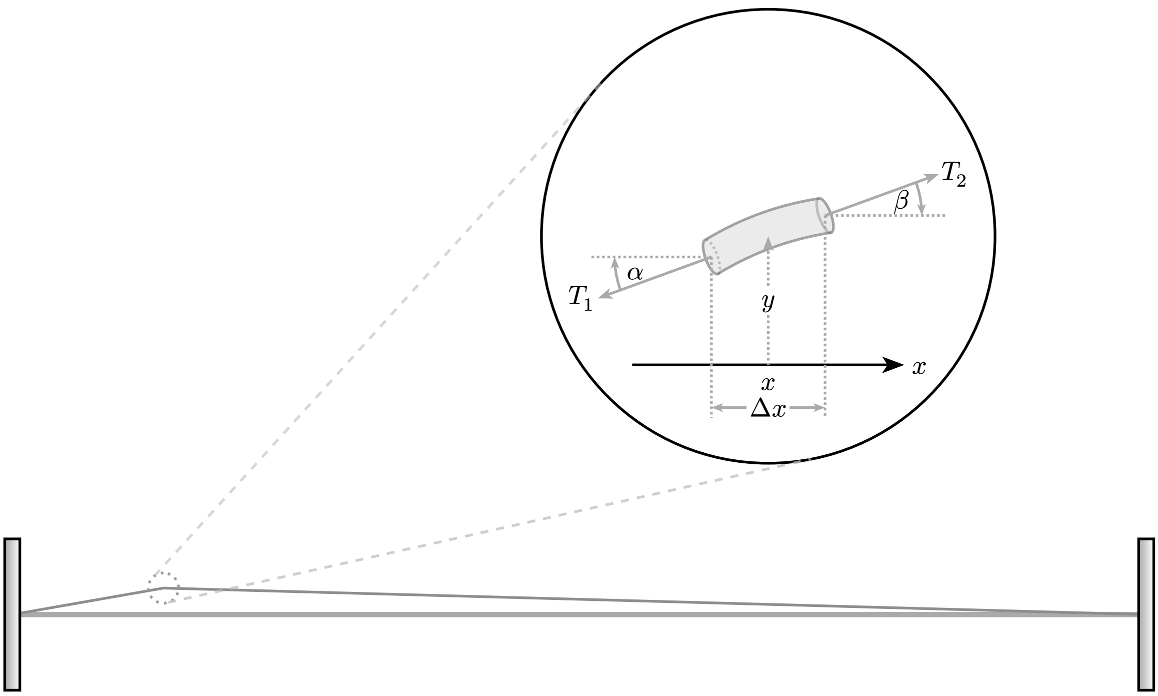
[illegible]

1 [illegible]

1. [illegible] 1 [illegible] $L(t)$ [illegible] $\mu(x,t)$ [illegible] $T(x,t)$ [illegible]
2. [illegible] $t=0$ [illegible] x_0 [illegible] A_0 [illegible]
3. [illegible] $y(x,t)$ [illegible] $f_y(x,t)$ [illegible]

2 [illegible]

1. [illegible] A_0 [illegible] $\frac{A_0}{L_0} < 1\%$?
2. [illegible] $L(t)$ [illegible] t [illegible] $T(x,t)$ [illegible] x [illegible] t [illegible]
3. [illegible]
   - [illegible] $L(t)$ [illegible] L_0
   - [illegible] $T(x,t)$ [illegible] T_0
4. [illegible] $T(x,t)$ [illegible] T_0 [illegible]
5. [illegible] x [illegible] $\mu(x,t)$ [illegible] μ_0 [illegible]
6. [illegible] $t=0$ [illegible]
7. [illegible]



[illegible] 1 [illegible]

3 [illegible]

3.1 [illegible]

[illegible] 1 [illegible] x [illegible] y [illegible]

3.2 [illegible]

[illegible]x[illegible]Δx[illegible]$\mu_0 \Delta x$[illegible]y[illegible]x[illegible]x[illegible]

[illegible]$T_2 \cos(\beta) - T_1 \cos(\alpha) = 0$

[illegible]

$$T_x = T_1 \cos(\alpha) = T_2 \cos(\beta) \tag{3.1}$$

[illegible]y[illegible]

$$T_2 \sin(\beta) - T_1 \sin(\alpha) = \mu_0 \Delta x \frac{\partial^2 y}{\partial t^2} \tag{3.2}$$

> [illegible]
>
> [illegible] Eq. (3.2) [illegible] $F = ma$[illegible] [illegible]$\frac{\partial^2 y}{\partial t^2}$ [illegible]

[illegible] Eq. (3.2) [illegible] T_x[illegible]

$$\frac{T_2 \sin(\beta)}{T_2 \cos(\beta)} - \frac{T_1 \sin(\alpha)}{T_1 \cos(\alpha)} = \frac{\mu_0 \Delta x}{T_x} \cdot \frac{\partial^2 y}{\partial t^2} \tag{3.3}$$

[illegible]

$$\frac{\tan(\beta) - \tan(\alpha)}{\Delta x} = \frac{\mu_0}{T_x} \cdot \frac{\partial^2 y}{\partial t^2} \tag{3.4}$$

[illegible]

$$\frac{\left.\frac{\partial y}{\partial x}\right|_{x+0.5\times\Delta x} - \left.\frac{\partial y}{\partial x}\right|_{x-0.5\times\Delta x}}{\Delta x} = \frac{\mu_0}{T_x} \cdot \frac{\partial^2 y}{\partial t^2} \tag{3.5}$$

[illegible] Eq. (3.5) [illegible] y [illegible] x [illegible]

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu_0}{T_x} \cdot \frac{\partial^2 y}{\partial t^2} \tag{3.6}$$

> **[illegible]T_x[illegible]T_0[illegible]**
>
> [illegible] Eq. (3.6) [illegible]μ_0[illegible]T_x[illegible]T_0[illegible]
>
> GPT-5.5 [illegible] Eq. (3.1) [illegible] x [illegible]T_x[illegible]T_x [illegible] T_0[illegible] $T_x \approx T_0$[illegible]

4 [illegible]

[illegible] Eq. (3.6) [illegible]

[illegible]

$$
\begin{aligned}
\frac{\partial^2 y}{\partial x^2} &= \frac{\mu_0}{T_0} \cdot \frac{\partial^2 y}{\partial t^2} \\
&= \frac{1}{c^2} \frac{\partial^2 y}{\partial t^2} \qquad c = \sqrt{\frac{T_0}{\mu_0}} \text{ [illegible]}
\end{aligned}
\tag{4.1}
$$

[illegible]

[illegible]$y(x,t)$[illegible]

$$y(x,t) = X(x) \cdot T(t) \tag{4.2}$$

[illegible]$T(t)$[illegible] T [illegible] t [illegible]-[illegible] t [illegible]

[illegible] x [illegible]$\frac{\partial y}{\partial x} = T(t)\frac{dX}{dx}$

[illegible] x [illegible]$\frac{\partial^2 y}{\partial x^2} = T(t)\frac{d^2X}{dx^2}$

[illegible]$\frac{\partial^2 y}{\partial t^2} = X(x) \cdot \frac{d^2T}{dt^2}$

[illegible] Eq. (4.1) [illegible]

$$T(t)\frac{d^2X}{dx^2} = \frac{1}{c^2} \cdot X(x)\frac{d^2T}{dt^2} \tag{4.3}$$

[illegible] Eq. (4.2) [illegible]

$$\frac{1}{X} \cdot \frac{d^2X}{dx^2} = \frac{1}{c^2} \cdot \frac{1}{T}\frac{d^2T}{dt^2} \tag{4.4}$$

[illegible] x [illegible] t [illegible] x [illegible] t [illegible] C[illegible] C [illegible]

[illegible] C [illegible]

[illegible] C [illegible] y(x,t)[illegible] X(x)[illegible] T(t)[illegible] C [illegible]

4.1 [illegible]

[illegible]

[illegible] Eq. (4.5) [illegible] Eq. (4.6) [illegible]

$$\frac{d^2X}{dx^2} - CX = 0 \tag{4.5}$$

$$\frac{d^2T}{dt^2} - C \cdot c^2 \cdot T = 0 \qquad \text{[illegible]} c = \sqrt{\frac{T_0}{\mu_0}} \tag{4.6}$$

4.2 [illegible]

[illegible]

$$\left.\begin{aligned} & y(0,t) = X(0)T(t) = 0 \\ & T(t)\text{[illegible]} \end{aligned}\right\} \implies X(0) = 0 \tag{4.7}$$

$$\left.\begin{aligned} & y(L_0,t) = X(L_0)T(t) = 0 \\ & T(t)\ \text{[illegible]} \end{aligned}\right\} \implies X(L_0) = 0 \tag{4.8}$$

4.3 [illegible]

[illegible] Eq. (4.5) [illegible]$r^2 = C$[illegible]$r_{1,2} = \pm\sqrt{C}$

[illegible] $C > 0$[illegible]$X(x) = C_1 e^{\sqrt{C}x} + C_2 e^{-\sqrt{C}x}$

[illegible] $C = 0$[illegible]$X(x) = C_1 + C_2 x$

[illegible] $C < 0$[illegible]$X(x) = C_1 \cos\left(\sqrt{-C}x\right) + C_2 \sin\left(\sqrt{-C}x\right)$

> **[illegible]**
>
> [illegible]C_1[illegible]C_2[illegible] X(x)[illegible] C [illegible]

[illegible] C	[illegible]	[illegible]	[illegible]	[illegible]
$C > 0$	$X(x) = C_1 e^{\sqrt{C}x} + C_2 e^{-\sqrt{C}x}$	$\left.\begin{aligned} & C_1+C_2=0 \\ & C_1\left(e^{\sqrt{C}L_0}-e^{-\sqrt{C}L_0}\right)=0 \\ & L_0>0 \end{aligned}\right\}$	$C_1 = C_2 = 0$	$X(x) = 0$
$C = 0$	$X(x) = C_1 + C_2 x$	$\left.\begin{aligned} & C_1=0 \\ & C_2 L_0=0 \\ & L_0>0 \end{aligned}\right\}$	$C_1 = C_2 = 0$	$X(x) = 0$
$C < 0$	$X(x) = C_1 \cos\left(\sqrt{-C}x\right) + C_2 \sin\left(\sqrt{-C}x\right)$	$\left.\begin{aligned} & C_1=0 \\ & C_2 \sin\left(\sqrt{-C}L_0\right)=0 \\ & L_0>0 \end{aligned}\right\}$	[illegible]$C = -k^2, k > 0$ [illegible]$k = \frac{n\pi}{L_0}$	[illegible]$X_n(x) = C_{2,n}\sin\left(\frac{n\pi}{L_0}x\right)$ [illegible]$C_{2,n}$[illegible] [illegible] $n \in \mathbb{N}^+$

[illegible]

$$X_n(x) = C_{2,n}\sin(k_n x) \qquad \text{[illegible]}\, k_n = \frac{n\pi}{L_0} \tag{4.9}$$

> **[illegible]**
>
> X(x)[illegible]
>
> $$X(x) = \sum_{n=1}^{\infty} C_{2,n}\sin(k_n x) \qquad \text{[illegible]}\, k_n = \frac{n\pi}{L_0} \tag{4.10}$$

4.4 [illegible]

$$\left.\begin{aligned} &\left.\frac{\partial y}{\partial t}\right|_{t=0} = X(x)\cdot \left.\frac{dT}{dt}\right|_{t=0} = 0 \\ &X(x) \not\equiv 0 \end{aligned}\right\} \quad \Longrightarrow \quad \left.\frac{dT}{dt}\right|_{t=0} = 0 \tag{4.11}$$

$$y(x,0) = X(x)T(0) = \begin{cases} \frac{A_0}{x_0}x & 0 \le x \le x_0 \\ \frac{A_0}{L_0 - x_0}(L_0 - x) & x_0 < x \le L_0 \end{cases} \tag{4.12}$$

4.5 [illegible]

[illegible] Eq. (4.6) [illegible]$r^2 - c^2 \cdot C = 0$[illegible] $C = -k_n^2, \quad k_n = \frac{n\pi}{L_0}$[illegible]

$$r_{1,2} = \pm i \cdot c \cdot k_n \tag{4.13}$$

[illegible]

$$T_n(t) = D_{1,n}\cos(\omega_n t) + D_{2,n}\sin(\omega_n t) \qquad \text{[illegible]}\omega_n = c \cdot k_n \tag{4.14}$$

[illegible] t [illegible]

$$\frac{dT_n}{dt} = D_{2,n}\omega_n \cos(\omega_n t) - D_{1,n}\omega_n \sin(\omega_n t) \tag{4.15}$$

[illegible] Eq. (4.11) [illegible]$D_{2,n} = 0$

[illegible]

$$T_n(t) = D_{1,n}\cos(\omega_n t) \qquad \text{[illegible]}\omega_n = c \cdot k_n \qquad k_n = \frac{n\pi}{L_0} \tag{4.16}$$

[illegible] $D_{1,n}$[illegible]$C_{2,n}$[illegible]

4.6 [illegible]

[illegible] $X_n(x) \cdot T_n(t) = C_{2,n}\sin(k_n x) \cdot D_{1,n}\cos(\omega_n t)$ [illegible] Eq. (4.1) [illegible] $y(x,t)$[illegible]

$$\begin{aligned} y(x,t) &= \sum_{n=1}^{\infty} X_n(x)T_n(t) \\ &= \sum_{n=1}^{\infty}\big(C_{2,n}D_{1,n}\sin(k_n x)\cos(\omega_n t)\big) \end{aligned} \tag{4.17}$$

[illegible] $k_n = \frac{n\pi}{L_0} \qquad \omega_n = c \cdot k_n \qquad n \in \mathbb{N}^+$

[illegible]

$$y(x,0) = \sum_{n=1}^{\infty}\left(C_{2,n}D_{1,n}\sin\left(\frac{n\pi}{L_0}x\right)\right) \tag{4.18}$$

[illegible] Eq. (4.12) [illegible]

$$\sum_{n=1}^{\infty}\left(C_{2,n}D_{1,n}\sin\left(\frac{n\pi x}{L_0}\right)\right)=\begin{cases}\frac{A_0}{x_0}x & 0\le x\le x_0\\ \frac{A_0}{L_0-x_0}(L_0-x) & x_0<x\le L_0\end{cases}\tag{4.19}$$

[illegible] $C_{2,n}D_{1,n}$ [illegible]

[illegible]

$$\begin{aligned}
b_n &= C_{2,n}D_{1,n}\\
&= \frac{2}{L_0}\int_0^{L_0} y(x,0)\sin\left(\frac{n\pi x}{L_0}\right)dx\\
&= \frac{2}{L_0}\left(\frac{A_0}{x_0}\int_0^{x_0} x\sin\left(\frac{n\pi x}{L_0}\right)dx+\frac{A_0L_0}{L_0-x_0}\int_{x_0}^{L_0}\sin\left(\frac{n\pi x}{L_0}\right)dx-\frac{A_0}{L_0-x_0}\int_{x_0}^{L_0}x\sin\left(\frac{n\pi x}{L_0}\right)dx\right)\\
&= \frac{2A_0}{L_0x_0}\int_0^{x_0}x\sin(k_nx)dx+\frac{2A_0}{L_0-x_0}\int_{x_0}^{L_0}\sin(k_nx)dx-\frac{2A_0}{L_0(L_0-x_0)}\int_{x_0}^{L_0}x\sin(k_nx)dx\\
&= -\frac{2A_0}{L_0x_0k_n}\left((x\cos(k_nx))\Big|_0^{x_0}-\int_0^{x_0}\cos(k_nx)dx\right)-\frac{2A_0}{(L_0-x_0)k_n}\cos(k_nx)\Big|_{x_0}^{L_0}+\frac{2A_0}{L_0(L_0-x_0)k_n}\left(x\cos(k_nx)\Big|_{x_0}^{L_0}-\int_{x_0}^{L_0}\cos(k_nx)dx\right)\\
&= -\frac{2A_0}{L_0x_0k_n}\left(x_0\cos(k_nx_0)-\frac{1}{k_n}\sin(k_nx_0)\right)-\frac{2A_0}{(L_0-x_0)k_n}(\cos(k_nL_0)-\cos(k_nx_0))\\
&+\frac{2A_0}{L_0(L_0-x_0)k_n}\left(L_0\cos(k_nL_0)-x_0\cos(k_nx_0)-\frac{1}{k_n}(\sin(k_nL_0)-\sin(k_nx_0))\right)\\
&= 2\left(-\frac{A_0}{L_0k_n}+\frac{A_0}{(L_0-x_0)k_n}-\frac{A_0x_0}{L_0(L_0-x_0)k_n}\right)\cos(k_nx_0)+2\left(-\frac{A_0}{(L_0-x_0)k_n}+\frac{A_0}{(L_0-x_0)k_n}\right)\cos(k_nL_0)\\
&+2\left(\frac{A_0}{L_0x_0k_n^2}+\frac{A_0}{L_0(L_0-x_0)k_n^2}\right)\sin(k_nx_0)-\frac{2A_0}{L_0(L_0-x_0)k_n^2}\sin(k_nL_0)\\
&= 0+0+\frac{2A_0}{L_0k_n^2}\cdot\frac{L_0}{x_0(L_0-x_0)}\sin(k_nx_0)-0\\
&= \frac{2A_0}{x_0(L_0-x_0)k_n^2}\sin(k_nx_0)\qquad \text{[illegible]}\,k_n=\frac{n\pi}{L_0}\\
&= \frac{2A_0L_0^2}{x_0(L_0-x_0)\pi^2n^2}\sin\left(\frac{n\pi x_0}{L_0}\right)
\end{aligned}\tag{4.20}$$

[illegible]

[illegible] $y(x,t)$ [illegible]

$$\begin{aligned}
y(x,t) &= \sum_{n=1}^{\infty}X_n(x)T_n(t)\\
&= \underbrace{\sum_{n=1}^{\infty}(b_n\sin(k_nx)\cos(\omega_nt))}_{\text{[illegible]}}\\
&= \frac{1}{2}\sum_{n=1}^{\infty}\left(b_n\left(\underbrace{\sin(k_nx+\omega_nt)}_{\text{[illegible]}}+\underbrace{\sin(k_nx-\omega_nt)}_{\text{[illegible]}}\right)\right)
\end{aligned}\tag{4.21}$$

[illegible] b_n [illegible] $\frac{1}{n^2}$ [illegible]

[illegible]

[illegible]

[illegible]

- [illegible] n [illegible] n [illegible]
  - $\sin(k_n x)$[illegible] n−1 [illegible]
  - $\cos(\omega_n t)$[illegible]ω_n[illegible]
  - b_n[illegible]
- [illegible]

[illegible]

[illegible]$\sin A \cos B = \frac{1}{2}(\sin(A+B) + \sin(A-B))$[illegible]

- $\sin(k_n x + \omega_n t)$[illegible]+ [illegible]
- $\sin(k_n x - \omega_n t)$[illegible]− [illegible]
- [illegible]

[illegible]

$$b_n = \frac{2A_0 L_0^2}{x_0(L_0 - x_0)\pi^2 n^2} \sin(k_n x_0) \qquad n \in \mathbb{N}^+ \tag{4.22}$$

$$k_n = \frac{n\pi}{L_0} \tag{4.23}$$

$$\omega_n = c \cdot k_n \tag{4.24}$$

[illegible]$k_n \lambda_n = 2\pi$[illegible] $\omega_n T_n = 2\pi$ [illegible]T_n[illegible]

[illegible]k_n[illegible] ω_n[illegible]$c = \sqrt{\frac{T_0}{\mu_0}}$[illegible]

[illegible]k_n [illegible]angular wave number[illegible]

5 [illegible]

[illegible]

- [illegible]
- [illegible])

	[illegible]	[illegible]
[illegible]	$X(x)$[illegible]	$T(t)$ [illegible] b_n[illegible]
[illegible]	[illegible] $t \geq 0$	[illegible] $t = 0$
[illegible]	[illegible] k_n [illegible] $\sin(k_n x)$	[illegible] b_n [illegible] $T_n(t)$
[illegible]	[illegible]	[illegible]

6 [illegible] $T_y(x,t)$

[illegible]

[illegible] 1 [illegible]

- [illegible]$\vec{T}(x,t) = |T(x,t)| \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$[illegible]
- [illegible]

$$
\begin{aligned}
T_y(x,t) &= \vec{T}(x,t) \cdot \hat{y} \\
&= |T(x,t)| \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
&= |T(x,t)| \sin\theta
\end{aligned}
\tag{6.1}
$$

- [illegible]
- [illegible] T_y [illegible] T_y [illegible] T_y [illegible]

[illegible]

$$
\begin{aligned}
T_y(x,t) &= |T(x,t)| \cdot \sin[\theta^1(x,t)] \\
&= |T(x,t)| \cdot \cos[\theta(x,t)] \cdot \tan[\theta(x,t)]
\end{aligned}
\tag{6.2}
$$

[illegible]

[illegible]

- [illegible] $\cos(\theta(x,t))$ [illegible] 1
- [illegible] $|T(x,t)|$[illegible]T_0

[illegible]

$$T_y(x,t) = T_0 \tan(\theta(x,t)) \tag{6.3}$$

$$T_y(x,t) = T_0 \cdot \frac{\partial y}{\partial x} \tag{6.4}$$

[illegible] Eq. (4.21) [illegible]

$$\frac{\partial y}{\partial x} = \frac{1}{2} \sum_{n=1}^{\infty} \left\{ b_n k_n \left[\cos(k_n x + \omega_n t) + \cos(k_n x - \omega_n t) \right] \right\} \tag{6.5}$$

[illegible]$T_y(x,t)$

$$T_y(x,t) = \frac{T_0}{2} \cdot \sum_{n=1}^{\infty} \left\{ b_n k_n \left[\underbrace{\cos(k_n x + \omega_n t)}_{\text{[illegible]}} + \underbrace{\cos(k_n x - \omega_n t)}_{\text{[illegible]}} \right] \right\} \tag{6.6}$$

[1] θ [illegible] $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$$
= T_0 \cdot \sum_{n=1}^{\infty}\left\{\underbrace{b_n k_n \cdot \cos(k_n x)\cos(\omega_n t)}_{\text{□□□□}}\right\} \tag{6.6}
$$

7 [illegible]

[illegible] 1 [illegible]

$$
\begin{aligned}
F_y &= T_2 \sin\beta - T_1 \sin\alpha \\
&= T_x(\tan\beta - \tan\alpha)
\end{aligned} \tag{7.1}
$$

[illegible] T_x [illegible]T_0[illegible]

$$
F_y \approx T_0 \frac{\partial^2 y}{\partial x^2}\Delta x \tag{7.2}
$$

[illegible]f_y

$$
\begin{aligned}
f_y &= \frac{F_y}{\Delta x} \\
&\approx T_0 \frac{\partial^2 y}{\partial x^2}
\end{aligned} \tag{7.3}
$$

[illegible] Eq. (4.21) [illegible]

$$
\frac{\partial y}{\partial x} = \frac{1}{2}\sum_{n=1}^{\infty}\left\{b_n k_n \left[\cos(k_n x + \omega_n t) + \cos(k_n x - \omega_n t)\right]\right\} \tag{7.4}
$$

[illegible]

$$
\begin{aligned}
\frac{\partial^2 y}{\partial x^2} &= -\frac{1}{2}\sum_{n=1}^{\infty}\left\{b_n k_n^2 \left[\sin(k_n x + \omega_n t) + \sin(k_n x - \omega_n t)\right]\right\} \\
&= -\sum_{n=1}^{\infty}\left\{b_n k_n^2 \cdot \sin(k_n x)\cos(\omega_n t)\right\}
\end{aligned} \tag{7.5}
$$

[illegible]

[illegible]f_y

$$
\begin{aligned}
f_y &\approx T_0 \frac{\partial^2 y}{\partial x^2} \\
&= -T_0 \sum_{n=1}^{\infty}\left\{\underbrace{b_n k_n^2 \cdot \sin(k_n x)\cos(\omega_n t)}_{\text{□□□□}}\right\} \\
&= -\frac{T_0}{2}\sum_{n=1}^{\infty}\left\{b_n k_n^2 \left[\underbrace{\sin(k_n x + \omega_n t)}_{\text{□□□}} + \underbrace{\sin(k_n x - \omega_n t)}_{\text{□□□}}\right]\right\}
\end{aligned} \tag{7.6}
$$

8 [illegible]

[illegible] t=0 [illegible]

[illegible]

$$
\begin{aligned}
\Delta K &= \frac{1}{2}\mu_0 \Delta x \left(\frac{\partial y}{\partial t}\right)^2 \\
&= \frac{\mu_0}{2}\left(\frac{\partial y}{\partial t}\right)^2 \Delta x
\end{aligned}
\tag{8.1}
$$

[illegible]

$$
\begin{aligned}
\Delta m &\approx \mu_0 \Delta s \\
&\approx \mu_0 \Delta x
\end{aligned}
\tag{8.2}
$$

- [illegible] μ_0[illegible]
- [illegible]Δs[illegible] Δx[illegible]

[illegible]Δs[illegible]Δx[illegible]$\Delta s - \Delta x \approx \frac{1}{2}\left(\frac{\partial y}{\partial x}\right)^2 \Delta x$

[illegible]

[illegible]

- [illegible] T_0 [illegible]
- [illegible] $\Delta s - \Delta x$[illegible] Δs [illegible]
- [illegible] $\times$ [illegible] $\times$ [illegible]

$$
\begin{aligned}
\Delta P &\approx T_0(\Delta s - \Delta x) \\
&\approx T_0\left(\sqrt{(\Delta x)^2 + (\Delta y)^2} - \Delta x\right) \\
&= T_0\left(\sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2} - 1\right)\Delta x
\end{aligned}
\tag{8.3}
$$

[illegible]

$$
\begin{aligned}
\sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2} - 1 &= \frac{1}{2}\left(\frac{\partial y}{\partial x}\right)^2 + O\left(\left(\frac{\partial y}{\partial x}\right)^4\right) \\
&\approx \frac{1}{2}\left(\frac{\partial y}{\partial x}\right)^2
\end{aligned}
$$

[illegible]

[illegible]

$$
\Delta P \approx \frac{T_0}{2}\left(\frac{\partial y}{\partial x}\right)^2 \Delta x
\tag{8.4}
$$

[illegible]

[illegible]

- [illegible] μ_0

[illegible]

- [illegible] T_0
- [illegible] $\Delta s = \sqrt{(\Delta x)^2 + (\Delta y)^2}$

[illegible]

[illegible]

$$E(t) = \int_0^{L_0} \left(\frac{\mu_0}{2} \left(\frac{\partial y}{\partial t} \right)^2 + \frac{T_0}{2} \left(\frac{\partial y}{\partial x} \right)^2 \right) dx \tag{8.5}$$

? [illegible]

[illegible]$x = x_0$[illegible]$\frac{\partial y}{\partial x}$[illegible]

[illegible] [illegible] $x = x_0$ [illegible]

[illegible] E(0) [illegible]

8.1 [illegible] $t = 0$ [illegible]

[illegible]$y(x,t)$[illegible]$t = 0$[illegible]$E(t) = E(0)$

[illegible]

$$\left. \frac{\partial y}{\partial t} \right|_{t=0} = 0 \tag{8.6}$$

$$\left. \frac{\partial y}{\partial x} \right|_{t=0} = \begin{cases} \frac{A_0}{x_0} & 0 \le x < x_0 \\ \text{□□□} & x = x_0 \\ \frac{A_0}{x_0 - L_0} & x_0 < x \le L_0 \end{cases} \tag{8.7}$$

[illegible]

$$\begin{aligned} E(0) &= \int_0^{x_0} \frac{T_0}{2} \frac{A_0^2}{x_0^2} dx + \int_{x_0}^{L_0} \frac{T_0}{2} \frac{A_0^2}{(L_0 - x_0)^2} dx \\ &= \frac{T_0 A_0^2 L_0}{2 x_0 (L_0 - x_0)} \end{aligned} \tag{8.8}$$

[illegible] $E(0)$ [illegible]

- [illegible]$E(0) \propto A_0^2$[illegible]LC [illegible]
- [illegible]$E(0) \propto T_0$[illegible]“[illegible]”[illegible]

- [illegible] x_0 [illegible] $x_0(L_0 - x_0)$[illegible]
  - [illegible] $x_0 \to 0$ [illegible] $x_0 \to L_0$[illegible]$E(0) \to \infty$[illegible] A_0 [illegible]——[illegible] $\frac{A_0}{x_0}$ [illegible]
  - [illegible] $x_0 = \frac{L_0}{2}$ [illegible] x_0 [illegible] $L_0 - x_0$ [illegible]
  - [illegible]$E(0)$ [illegible]

$$E(0) = \frac{T_0}{2}\left[\underbrace{\left(\frac{A_0}{x_0}\right)^2 \cdot x_0}_{\text{[illegible]}} + \underbrace{\left(\frac{A_0}{L_0 - x_0}\right)^2 \cdot (L_0 - x_0)}_{\text{[illegible]}}\right] \tag{8.9}$$

[illegible] = $\frac{T_0}{2} \cdot (\text{[illegible]})^2 \cdot$ [illegible]

9 [illegible]

[illegible]

9.1 [illegible]

[illegible]

1. [illegible] $\frac{A_0}{L_0} \ll 1$[illegible] 1%[illegible] $\left|\frac{\partial y}{\partial x}\right| \ll 1$[illegible]
2. [illegible] $L(t)$ [illegible] L_0[illegible]
3. [illegible] $T(x,t)$ [illegible] T_0[illegible]
4. [illegible] $\mu(x,t)$ [illegible] μ_0[illegible]
5. [illegible]
6. [illegible] $t = 0$ [illegible]
7. [illegible]
   - [illegible] x [illegible] T_x [illegible]
   - [illegible] T_x [illegible] T_0[illegible]
   - [illegible] $\cos\theta \approx 1$[illegible]
8. [illegible]
   - [illegible] Δs [illegible] Δx[illegible]
   - [illegible] $\sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2} - 1$ [illegible] $\frac{1}{2}\left(\frac{\partial y}{\partial x}\right)^2$[illegible] $O\left(\left(\frac{\partial y}{\partial x}\right)^4\right)$ [illegible]

[illegible] N_{terms} [illegible]

9.2 [illegible]

[illegible]

[illegible]

[illegible]

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 y}{\partial t^2} \quad \left(c = \sqrt{\frac{T_0}{\mu_0}}\right) \tag{9.1}$$

[illegible]

9.3 [illegible]

[illegible]

1. [illegible] $\varepsilon = \frac{A_0}{L_0}$ [illegible] $\varepsilon < 0.01$ [illegible] 1[illegible]
2. [illegible] 5%[illegible]
3. [illegible] $\frac{\Delta T}{T_0} \sim \varepsilon^2$???[illegible]
4. [illegible] $A_0 \to 0$ [illegible]
5. [illegible] ε [illegible]

9.4 [illegible]

[illegible]

1. [illegible] $O\left(\left(\frac{\partial y}{\partial x}\right)^4\right)$ [illegible] $O(\varepsilon^2)$[illegible] $\varepsilon = 0.01$ [illegible] 10^{-4} [illegible]
2. [illegible] ε[illegible] $y = y_0 + \varepsilon y_1 + \varepsilon^2 y_2 + \ldots$[illegible] y_0 [illegible]y_1, y_2 [illegible]
3. [illegible] $E(t) \equiv E(0)$[illegible] $E(t) - E(0)$ [illegible]
4. [illegible]
5. [illegible] ε^2[illegible] $\varepsilon < 0.05$ [illegible] 1%[illegible]

9.5 [illegible]

[illegible]

1. [illegible] μ_0 [illegible]
2. [illegible] $|\frac{\partial y}{\partial x}| \ll 1$[illegible]
3. [illegible] T_0 [illegible]
4. [illegible]
5. [illegible]
6. [illegible]

[illegible]

9.6 [illegible]

[illegible] $\frac{A_0}{L_0} < 0.02$ [illegible]

10 [illegible]

[illegible] $\frac{A_0}{L_0} \ll 1$[illegible]

10.1 [illegible] ε [illegible]

[illegible]

$$\varepsilon = \frac{A_0}{L_0} \tag{10.1}$$

[illegible]

$$\varepsilon = \frac{0.01}{0.6} = \frac{1}{60} \approx 0.0167 \tag{10.2}$$

[illegible] $\frac{A_0}{L_0} < 0.02$ [illegible]

10.2 [illegible]

[illegible]

$$m = \max\left(\frac{A_0}{x_0}, \frac{A_0}{L_0 - x_0}\right) \tag{10.3}$$

[illegible] $x_{\text{rel}} = 0.1$[illegible] $x_0 = 0.06$ m[illegible]

$$m = \frac{0.01}{0.06} \approx 0.1667 = 10\varepsilon \tag{10.4}$$

[illegible]

$$\theta_{\max} = \arctan(m) \approx 0.165 \text{ rad} \approx 9.46^\circ \tag{10.5}$$

[illegible]

$$\cos\theta_{\max} \approx 0.986 \tag{10.6}$$

[illegible] 1.4%[illegible] $\cos\theta \approx 1$ [illegible]<2%[illegible]

[illegible] x_{rel} [illegible] 0.5 [illegible]$m \approx 2\varepsilon \approx 0.033$[illegible]$\theta_{\max}$ [illegible]

[illegible]θ[illegible] $\theta \approx m$ [illegible]$\lim_{\theta \to 0} \frac{\tan\theta}{\theta} = 1$ [illegible] $\theta \approx \tan\theta$ [illegible]

10.3 [illegible]

[illegible]

$$\sqrt{1 + s^2} - 1 = \frac{1}{2}s^2 + R_4(s) \tag{10.7}$$

[illegible] $R_4(s) = O(s^4)$[illegible] $O(s^2)$[illegible]

[illegible] $s = m = 10\varepsilon$[illegible]

$$O(s^2) = O(100\varepsilon^2) \approx 100 \times (0.0167)^2 \approx 0.0279 \quad (2.8\%) \tag{10.8}$$

[illegible]$m \approx 2\varepsilon$[illegible]

$$O(4\varepsilon^2) \approx 0.0011 \quad (0.11\%) \tag{10.9}$$

[illegible]$x_0 \to 0$ [illegible] L_0[illegible] ε^2 [illegible]

10.4 [illegible] $T_x \approx T_0$ [illegible]

[illegible] T_x [illegible]

$$\begin{aligned} |T| &= \frac{T_x}{\cos\theta} \\ &\approx \frac{T_x}{1 - \frac{1}{2}\theta^2 + O(\theta^4)} \\ &\approx \frac{T_x\left[1 - \frac{1}{2}\theta^2 + O(\theta^4) + \frac{1}{2}\theta^2 - O(\theta^4)\right]}{1 - \frac{1}{2}\theta^2 + O(\theta^4)} \\ &\approx T_x\left[1 + O(\theta^2)\right] \end{aligned} \tag{10.10}$$

[illegible]

$$\frac{\Delta\,|T|}{T_x} = O(\theta^2) = O(m^2) = O(100\varepsilon^2) \approx 0.0279 \quad (2.8\%) \tag{10.11}$$

[illegible] $|T(x,t)| \approx T_0$ [illegible] 3%[illegible] ε [illegible]

10.5 [illegible]

[illegible]

$$\Delta L \approx \int_0^{L_0} \frac{1}{2}\left(\frac{\partial y}{\partial x}\right)^2 dx \approx \frac{1}{2}m^2 \cdot \left(\frac{L_0}{2}\right) = O(\varepsilon^2 L_0) \tag{10.12}$$

$$\Delta L \approx \int_0^{L_0} \frac{1}{2}\left(\frac{\partial y}{\partial x}\right)^2 dx < \frac{1}{2}m^2 \cdot L_0 = O(100\varepsilon^2 L_0) \tag{10.13}$$

[illegible]

$$\frac{\Delta L}{L_0} = O(100\varepsilon^2) \approx 0.028 \quad (2.8\%) \tag{10.14}$$

[illegible] $O(\varepsilon^2)$ [illegible]

[illegible] $\mu(x,t) \approx \mu_0$ [illegible] $L(t) \approx L_0$[illegible]

10.6 [illegible]

[illegible]

$$c = \sqrt{\frac{T_0}{\mu_0}} = \sqrt{\frac{80}{0.01}} = \sqrt{8000} \approx 89.44 \text{ m/s} \tag{10.15}$$

[illegible]

$$f_1 = \frac{c}{2L_0} \approx \frac{89.44}{1.2} \approx 74.53 \text{ Hz} \tag{10.16}$$

$$T_1 = \frac{1}{f_1} \approx 0.0134 \text{ s} \tag{10.17}$$

[illegible] $cT_1 = 1.2$ m[illegible] $2L_0$[illegible]

10.7 [illegible]

[illegible] $1\% \sim 3\%$ [illegible]

$$\varepsilon < 0.01 \quad [illegible] \quad x_{\text{rel}} \in [0.2, 0.8] \tag{10.18}$$

[illegible] $m < 5\varepsilon$[illegible] 0.5% [illegible]

11 [illegible]$y(x,t)$[illegible]

[illegible]$y(x,t)$[illegible] N [illegible] $y(x,t)$ [illegible]

11.1 [illegible]

[illegible] $y(x,t)$[illegible] N [illegible] $y_N(x,t)$[illegible]

$$y_N(x,t) = \sum_{n=1}^{N} b_n \sin(k_n x)\cos(\omega_n t) \tag{11.1}$$

[illegible]

$$\begin{aligned} E_N(x,t) &= y(x,t) - y_N(x,t) \\ &= \sum_{n=N+1}^{\infty} b_n \sin(k_n x)\cos(\omega_n t) \end{aligned} \tag{11.2}$$

11.2 [illegible]

[illegible] x [illegible] t [illegible] $|\sin(k_n x)\cos(\omega_n t)| \le 1$[illegible]:

$$|E_N(x,t)| \le \sum_{n=N+1}^{\infty} |b_n| \tag{11.3}$$

[illegible]

$$b_n = \frac{2A_0L_0^2}{x_0(L_0 - x_0)\pi^2 n^2}\sin(k_n x_0) \tag{11.4}$$

[illegible] $|\sin(k_n x_0)| \le 1$[illegible]

$$|b_n| \le \frac{2A_0L_0^2}{x_0(L_0 - x_0)\pi^2} \cdot \frac{1}{n^2} \equiv \frac{C}{n^2} \tag{11.5}$$

[illegible] C [illegible]

$$C = \frac{2A_0 L_0^2}{x_0(L_0 - x_0)\pi^2} \tag{11.6}$$

[illegible]

$$|E_N(x,t)| \le C \sum_{n=N+1}^{\infty} \frac{1}{n^2} \tag{11.7}$$

[illegible] $\frac{1}{x^2}$ [illegible] $\int_{N+1}^{\infty} \frac{1}{x^2} dx < \sum_{n=N+1}^{\infty} \frac{1}{n^2} < \int_N^{\infty} \frac{1}{x^2} dx$ [illegible]

$$\sum_{n=N+1}^{\infty} \frac{1}{n^2} < \int_N^{\infty} \frac{1}{x^2} dx = \frac{1}{N} \tag{11.8}$$

[illegible]

$$|E_N(x,t)| \le \frac{C}{N} \tag{11.9}$$

11.3 [illegible]

[illegible] $\varepsilon_N \le \frac{C}{N}$ [illegible]

11.3.1 [illegible] $|\sin(k_n x_0)| \le 1$

[illegible] b_n [illegible] $\sin(k_n x_0) = \sin(n\pi x_{\text{rel}})$ [illegible] 1 [illegible] n [illegible]—— [illegible] $|b_n|$ [illegible]

[illegible] $x_{\text{rel}} = 0.1$ [illegible] $n > N = 50$ [illegible] $\sin(k_n x_0) = \sin(0.1\pi n)$ [illegible] 20 [illegible]

n	$0.1\pi n$	$\sin(0.1\pi n)$
51	5.1π	-0.309
55	5.5π	-1
60	6π	0
65	6.5π	1
70	7π	0

[illegible] 20 [illegible] 2 [illegible] $|\sin| = 1$ [illegible] 4 [illegible] 14 [illegible] 0 [illegible] 1 [illegible] N [illegible] $|\sin(k_n x_0)|$ [illegible] $\frac{2}{\pi} \approx 0.637$ [illegible] $|b_n|$ [illegible] $\frac{1}{0.637} - 1 \approx 57\%$ [illegible]

[illegible] $x_{\text{rel}} = 0.5$ [illegible] $\sin(k_n x_0) = \sin(n\frac{\pi}{2})$ [illegible] n [illegible]

11.3.2 [illegible] $|\sin(k_n x)\cos(\omega_n t)| \le 1$

[illegible] n [illegible] (x,t) [illegible] $b_n \sin(k_n x)\cos(\omega_n t)$ [illegible] $|\sin(k_n x)\cos(\omega_n t)| \le 1$ [illegible] (x,t) [illegible] 1 [illegible]

1. [illegible] $|\sin(k_n x)\cos(\omega_n t)| = 1$ [illegible] $|\sin(k_n x)| = 1$ [illegible] $|\cos(\omega_n t)| = 1$ [illegible] $k_n x = \frac{\pi}{2} + m\pi$ [illegible] $\omega_n t = l\pi$ [illegible] (x,t) [illegible] n [illegible]

2. [illegible] 1 [illegible] $\sum b_n \sin(k_n x)\cos(\omega_n t)$ [illegible] $\sum |b_n|$ [illegible]

3. [illegible] n [illegible] $\sin(k_n x)$ [illegible] x [illegible] n [illegible] $\cos(\omega_n t)$ [illegible] t [illegible] n [illegible]

11.3.2.1 [illegible]

RMS [illegible] Root Mean Square [illegible] RMS [illegible] “[illegible]” [illegible]

1. [illegible]**Square**[illegible]
2. [illegible]**Mean**[illegible]
3. [illegible]**Root**[illegible]

[illegible] $[a, b]$ [illegible] $f(x)$[illegible] RMS [illegible]

$$f_{\text{RMS}} = \sqrt{\frac{1}{b-a} \int_a^b [f(x)]^2 dx} \tag{11.10}$$

[illegible] $x_1, x_2, \ldots, x_N$[illegible]RMS [illegible]

$$x_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} x_i^2} \tag{11.11}$$

[illegible] $\langle \cdot \rangle$ [illegible] $\langle \cdot \rangle$ [illegible]mean[illegible] $\langle f \rangle = \frac{1}{[illegible]} \int_{[illegible]} f d([illegible])$[illegible]

- $\langle \sin^2(k_n x) \rangle$ [illegible] $\sin^2(k_n x)$ [illegible] $x \in [0, L_0]$[illegible] $\left[0, \frac{\pi}{k_n}\right]$[illegible]
- $\langle \cos^2(\omega_n t) \rangle$ [illegible] $\cos^2(\omega_n t)$ [illegible] $\left[0, \frac{\pi}{\omega_n}\right]$[illegible]

[illegible]

[illegible] $\boldsymbol{\langle \sin^2 \theta \rangle = \frac{1}{2}}$[illegible] $\sin^2 \theta = \frac{1-\cos(2\theta)}{2}$[illegible] θ [illegible] $[0, \pi]$ [illegible]

$$\langle \sin^2 \theta \rangle = \frac{1}{\pi} \int_0^\pi \sin^2 \theta d\theta = \frac{1}{\pi} \int_0^\pi \frac{1-\cos(2\theta)}{2} d\theta = \frac{1}{2} \tag{11.12}$$

[illegible] $\cos(2\theta)$ [illegible] $\frac{1}{2}$[illegible] $\langle \cos^2 \theta \rangle = \frac{1}{2}$[illegible]

[illegible] $\sin(k_n x)$ [illegible] x[illegible]$\cos(\omega_n t)$ [illegible] t[illegible] x [illegible] t [illegible]

$$\langle \sin^2(k_n x) \cos^2(\omega_n t) \rangle = \langle \sin^2(k_n x) \rangle \cdot \langle \cos^2(\omega_n t) \rangle \tag{11.13}$$

[illegible] RMS[illegible]

$$\text{RMS} = \sqrt{\langle \sin^2(k_n x) \cos^2(\omega_n t) \rangle} = \sqrt{\langle \sin^2(k_n x) \rangle \cdot \langle \cos^2(\omega_n t) \rangle} = \sqrt{\frac{1}{2} \times \frac{1}{2}} = \frac{1}{2} \tag{11.14}$$

[illegible] $\langle \cdot \rangle$ [illegible] x [illegible] t [illegible] $\frac{1}{2}$[illegible] 1[illegible] $|b_n|$ [illegible] 2 [illegible]

11.3.2.2 [illegible]
[illegible]

$$\sin(k_n x) \cos(\omega_n t) = \frac{1}{2}[\sin(k_n x + \omega_n t) + \sin(k_n x - \omega_n t)] \tag{11.15}$$

[illegible] (x, t) [illegible] n [illegible] $\sin(k_n x \pm \omega_n t)$ [illegible] $(k_n x \pm \omega_n t) = n\pi\left(\frac{x}{L_0} \pm c\frac{t}{L_0}\right)$ [illegible] n [illegible] n [illegible]

11.3.2.3 [illegible]
[illegible] $L_0 = 0.6$[illegible]$x_{\text{rel}} = 0.1$[illegible]$t = 0.005$ s [illegible] $n = 51$ [illegible] 55 [illegible] $x = 0.3$ m [illegible]

n	$\sin(k_n x) \cos(\omega_n t)$	$\lvert \sin(k_n x) \cos(\omega_n t) \rvert$	[illegible]

51	-0.165	0.165	1
52	0.035	0.035	1
53	0.203	0.203	1
54	-0.201	0.201	1
55	-0.095	0.095	1
[illegible]	-0.223	0.699	5

[illegible] 5[illegible] -0.223[illegible] 0.699[illegible]

- [illegible] 4.5%[illegible]$\frac{0.223}{5}$[illegible]
- [illegible] 14%[illegible]$\frac{0.699}{5}$[illegible]

[illegible]$|\sin(k_n x)\cos(\omega_n t)| \leq 1$ [illegible] 5 [illegible] 20 [illegible] (x, t) [illegible]

11.3.2.4 [illegible]

[illegible] 4 [illegible] 10 [illegible]

- [illegible] RMS [illegible] 1 [illegible] 2 [illegible]
- [illegible] 2 [illegible] 5 [illegible] (x, t) [illegible]

[illegible] 1.57 [illegible] 1.005 [illegible] (x, t) [illegible]——[illegible]

11.3.3 [illegible] $\sum_{n=N+1}^{\infty} \frac{1}{n^2} < \frac{1}{N}$

[illegible] $\frac{1}{N}$ [illegible]

$$\sum_{n=N+1}^{\infty} \frac{1}{n^2} < \int_N^{\infty} \frac{1}{x^2} dx = \frac{1}{N} \tag{11.16}$$

[illegible] Euler-Maclaurin [illegible]

$$\sum_{n=N+1}^{\infty} \frac{1}{n^2} = \frac{1}{N} - \frac{1}{2N^2} + O\left(\frac{1}{N^3}\right) \tag{11.17}$$

[illegible] $N = 50$[illegible]

- [illegible] $\frac{1}{N} = 0.02000$
- [illegible] $\sum_{n=51}^{\infty} \frac{1}{n^2} \approx 0.01990$
- [illegible] 0.5%

[illegible]

11.3.4 [illegible]

[illegible]

[illegible]	[illegible]	[illegible]
$\lvert\sin(k_n x_0)\rvert \leq 1$	[illegible] 57%[illegible]	[illegible] N [illegible]
$\lvert\sin(k_n x)\cos(\omega_n t)\rvert \leq 1$	[illegible] 4 [illegible] 10 [illegible]RMS + [illegible]	[illegible] (x, t) [illegible]
[illegible] $\frac{1}{N}$	[illegible] 0.5%[illegible]	[illegible] N [illegible]

[illegible] $|E_N(x,t)|$ [illegible] $\frac{C}{N}$ [illegible] $\frac{1}{5}$ [illegible] $\frac{1}{3}$[illegible] RMS [illegible] $\frac{1}{2}$ [illegible] x_{rel} [illegible] 0.5[illegible]t [illegible] $\frac{1}{10}$ [illegible]

[illegible] 4.5% [illegible] 2% [illegible] $\varepsilon \approx 1.67\%$ [illegible]

11.4 [illegible]

[illegible] ε_N [illegible] $x_0(L_0 - x_0)$[illegible]

- [illegible] $x_0 \to 0$ [illegible] $x_0 \to L_0$[illegible]$\varepsilon_N \to \infty$——[illegible]
- [illegible] $x_0 = \frac{L_0}{2}$[illegible]ε_N [illegible] $\sin(k_n x_0) = \sin(\frac{n\pi}{2})$ [illegible] n [illegible]

11.5 [illegible]

"[illegible]"[illegible]sharp bound[illegible] (x,t) [illegible] $|E_N(x,t)| \leq \frac{C}{N}$ [illegible]

1. [illegible] $|\sin(k_n x_0)| = 1$ [illegible] $n \geq N+1$ [illegible] $k_n x_0 = \frac{\pi}{2} + m_n \pi$[illegible] $\frac{x_0}{L_0} = \frac{1+2m_n}{2n}$ [illegible] n [illegible]——[illegible]
2. [illegible] $|\sin(k_n x)\cos(\omega_n t)| = 1$ [illegible]——[illegible] n [illegible]
3. [illegible] $|E_{N(x,t)}|$ [illegible] $\sum |b_n|$[illegible]

[illegible] $\frac{C}{N}$ [illegible]——[illegible]

11.6 [illegible]

[illegible] $N_{\text{terms}} = 50$ [illegible]

[illegible]	[illegible]	[illegible]
[illegible]	4.5%	$\approx 2\%$
[illegible]	$\frac{1}{N} \approx 2\%$	$\approx 1.2\%$

[illegible] $\varepsilon \approx 1.67\%$ [illegible] $\varepsilon_N \propto \frac{1}{N}$ [illegible] $N = 50$ [illegible] $N = 100$ [illegible]

12 [illegible]$y(x,t)$[illegible]

12.0.1 [illegible]

[illegible] y_N [illegible] y [illegible]

$$e_N^{\text{strict}}(x,t) = \frac{|E_N(x,t)|}{|y(x,t)|}, \quad \varepsilon_N^{\text{strict}} = \max_{x \in [0,L_0], t \geq 0} e_N^{\text{strict}}(x,t) \tag{12.1}$$

[illegible]

1. [illegible] $y(x,t)$ [illegible] $|y(x,t)|$ [illegible]
2. [illegible]$y(x,t) \approx 0$[illegible] $|y(x,t)|$ [illegible] $|E_{N(x,t)}|$ [illegible] $x = 0, L_0$ [illegible] $\sin(k_n x) = 0$[illegible] e_N^{strict} [illegible]——[illegible]

[illegible] A_0 [illegible] $|y(x,t)|$ [illegible]

$$
\begin{aligned}
\varepsilon_N &= \max_{x\in[0,L_0],t\geq 0} \frac{|E_N(x,t)|}{A_0} \\
&\leq \frac{C}{NA_0} \\
&= \frac{2L_0^2}{x_0(L_0-x_0)\pi^2 N}
\end{aligned}
\tag{12.2}
$$

12.0.2 [illegible]

[illegible]$L_0 = 0.6\ \mathrm{m}, x_{\mathrm{rel}} = 0.1, x_0 = 0.06\ \mathrm{m}, N_{\mathrm{terms}} = 50$[illegible]

$$
\varepsilon_N \leq \frac{2\times 0.6^2}{0.06\times 0.54\times \pi^2\times 50} = \frac{0.72}{0.0324\times 9.8696\times 50} = \frac{0.72}{15.99} \approx 4.5\% \tag{12.3}
$$

[illegible]

- $|\sin(k_n x_0)| \leq 1$[illegible] $\sin(k_n x_0)$ [illegible] n [illegible] 1[illegible]
- $|\sin(k_n x)\cos(\omega_n t)| \leq 1$[illegible] (x,t) [illegible]
- [illegible] $\sum \frac{1}{n^2} \leq \frac{1}{N}$ [illegible] N [illegible]

[illegible] $\frac{1}{4}$ [illegible] $\frac{1}{2}$[illegible] 1% [illegible] 2%[illegible] $\varepsilon = \frac{A_0}{L_0} \approx 1.67\%$ [illegible]

12.1 [illegible]

[illegible] A_0 [illegible]

1. [illegible] $|y(x,t)| \leq A_0$ [illegible] (x,t) [illegible]

$$
e_N^{\mathrm{strict}}(x,t) = \frac{|E_N(x,t)|}{|y(x,t)|} \geq \frac{|E_N(x,t)|}{\max|y(x,t)|} \geq \frac{|E_N(x,t)|}{A_0} \tag{12.4}
$$

[illegible] A_0 [illegible]——[illegible]

1. [illegible] $|y(x,t)| \approx A_0$ [illegible] $t=0$ [illegible]

2. [illegible]A_0 [illegible] $\max|y(x,t)|$ [illegible]

[illegible] ε_N [illegible]